

Million Merid Afessa (Ph.D)

Assistant Professor

Phone No. | +251913205174|

Personal Email: millionmerid208@gmail.com|

Institutional Email: | million.merid@ju.edu.et|

Google Talk | millionmerid208@gmail.com |

WhatsApp No. |+251913205174|

[Orcid ID: http://orcid.org/0000-0003-1598-0139](http://orcid.org/0000-0003-1598-0139)

<https://www.linkedin.com/feed/>

[Million M. Afessa - Google Scholar](#)

SHORT BIO

I am an Assistant Professor of Mechanical Engineering at Jimma Institute of Technology, Jimma University, Ethiopia, with a specialization in Sustainable Energy Engineering. With over 14 years of academic, research, training, and consulting experience. I hold a PhD in Sustainable Energy Engineering (2019–2025), jointly pursued at Politecnico di Milano (Department of Chemistry, Materials, and Chemical Engineering – CMIC) and Jimma Institute of Technology (JiT), Jimma University. My PhD research focuses on modeling and experimental investigation of the pyrolysis of multi-scale lignocellulosic biomass particles for biofuels application from coffee husk and khat stem valorization. I earned my MSc in Renewable Energy Technologies (2011–2014) from Addis Ababa Institute of Technology, Addis Ababa University, and BSc in Mechanical Engineering (2007–2010) from Mekelle University. My research interests include biomass conversion for biofuels applications, Sustainable Aviation Fuels (SAF), Green Hydrogen, bioenergy systems, pyrolysis, biochar production and utilization, low-carbon technologies for sustainable energy, including LCA & Techno-Economic Analysis. I am particularly committed to advancing circular economic solutions through the reuse of biobased waste streams. Beyond my academic roles, I actively lead and managed community-based projects promoting clean energy access, youth engagement, and safeguard environmental sustainability.

EDUCATION

Jimma University, JU and
Politecnico Di Milano,
Jan 2019 – Mar 2025

PhD in Sustainable Energy Engineering

Research Topic: Modeling and Experimental Investigation on the Pyrolysis of Multi-Scale Lignocellulosic Biomass Particles for Coffee Husk and Khat Stem Valorization.

Addis Ababa Institute of
Technology (AAIT),
Addis Ababa University
Sep 2011 – Apr 2014
Mekelle University,
Ethiopia
Sep 2007 – Jul 2010

MSc in Renewable Energy Technologies

Research Topic: Prediction of Performance Parameters and Determination of Aerodynamic Characteristics of Wind Turbine Airfoil Using CFD Modelling: A Case Study of Adama I Wind Turbine Blade Airfoil Section.

BSc in Mechanical Engineering

Research Topic: Design of Solar Greenhouse for the Application of Flower Industry in Ethiopia.

WORK EXPERIENCE

Jimma Institute of
Technology, Ethiopia
(**Sep 2025 – Presently**).

Director , Registrar and Student Affairs, Jimma Institute of Technology, Jimma University

- Act as a bridge between faculty, administration, and students on academic matters.
- Oversee student admission processes, course registration, and enrollment verification.
- Address student needs, grievances, and counseling services.
- Provide guidance on internships, job placements, and career development.

Jimma Institute of
Technology, Jimma
University, Ethiopia
(**Mar 2025 – Presently**).

Assistant Professor of Mechanical Engineering

- Encompass a combination of academic, research, Mentorship, community service, and institutional development responsibilities.

Senior Lecturer and Researcher

- Lead research on pyrolysis, gasification, and biofuels.
- Supervise students and mentor researchers.
- Teach and train in energy and technology.
- Support community energy initiatives.

Jimma Institute of
Technology, Jimma
University, Ethiopia
(**Apr 2014 – Mar 2025**)

Guest Researcher

- Conducted multiscale pyrolysis and kinetic modeling of biomass process biofuel research
- Developed models for biomass decomposition
- Published in top journals and conferences

Politecnico di Milano,
CRECK Research Lab
(**Jan 2019 – July 2019 and
Oct 2022 - Oct 2023**)

Project Evaluation & Proposal development Expert (Technical & Financial)

AYU sustainable energy
engineering works association
(**Oct2022 – Nov 2024**).

- Prepare winning technical and financial proposals
- Secure funding for sustainable energy initiatives
- Develop and implement clean energy projects
- Lead project execution and delivery

Jimma Institute of
Technology, Jimma University,
Ethiopia
(Oct 2010 – Apr 2014)

Assistant Lecturer II

- Taught undergraduate courses, and conducted labs sessions
- Assisted in curriculum development.
- Supervised undergraduate research projects and assisted academic activities.

Project Coordinator, ENERGISE Initiative

- Project Coordinator for the ENERGISE project at Jimma University
- Focused on sustainable energy engineering and local development impact

Senior Clean Cooking Expert (Based on Part time)

- Partnering with local organizations, women's groups, and businesses to facilitate product adoption and collect user feedback.
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Master Trainer and Training Facilitator

- Led impactful trainings on employability, life skills, and youth empowerment
- Championed leadership and positive change in social and environmental issues

Master Trainer at the Inspired Ethiopian Youth Association

- Delivered training on sustainable energy, climate change, and community development

Jimma Institute of
Technology, Jimma University,
Ethiopia
(Sep 2015 – Jan 2017)

Ethiopian Clean Cooking
Alliance (ECCA)
(October 2023 – Present)

Jimma Institute of
Technology, Jimma University,
Ethiopia
(Sep 2022 - Present).
Inspired Ethiopian Youth
Association
Jun 2024 - Aug 2024

SPONSORED RESEARCH / PROJECTS

Jimma University
2024 - 2026
(on progress)

Sustainable biofuel production and value-added products from waste resources: climate change mitigation, enhancing energy and food security with Circular Economy (Bio-based Circular Economy) .

Jimma University
2016 -2017
(Completed)

Renewable Energy for Sustainable Life: Techno- Economic Feasibility of Energy supply of stand-alone PV-Wind- Biomass Hybrid System for Rural Electrification: A case study in Jimma Zone.

Jimma University
2014- 2016
(Completed)

Potential Assessment, Design, and Implementation of Biogas System from Dining Hall and Kitchen Wastes at Jimma Institute of Technology, Jimma University.

LIST OF PUBLICATIONS AND CONFERENCE PARTICIPATION

- Teka Tesfaye Mengesha, Venkata Ramayya Ancha, Abebe Nigussie, **Million M. Afessa** and Ramchandra Bhandari Effect of Particle Size and Heating Rate on Formation of Polycyclic Aromatic Hydrocarbons During Corn Cob Biomass Pyrolysis, Sustainability 2025, 17(11), 4962; <https://doi.org/10.3390/su17114962> - 28 May 2025.
- **Afessa, M.M. et al. (2025)** 'Pyrolysis of large biomass particles: model validation and application to Coffee Husks valorization', Journal of Analytical and Applied Pyrolysis, p. 107028. Available at: <https://doi.org/10.1016/j.jaap.2025.107028>.
- **Afessa, M.M., Olu, F.E., Geleta, W.S. et al.** Unlocking the potential of biochar derived from coffee husk and khat stem for catalytic tar cracking during biomass pyrolysis: characterization and evaluation. Biomass Conv. Bioref. (2024) <https://doi.org/10.1007/s13399-024-05957-9>.
- Mekonnen Mossissa, H., **Afessa, M. M.**, T. Nega, D., Ahsan Amjed, M., & Venkata Ramayya, A. (2023). Opportunities and Challenges of Harnessing Biomass Wastes for Decentralized Heat and Energy Generation and Climate Mitigation via Fluidizedbed Gasification Pathway. IntechOpen [DOI: 10.5772/intechopen.112533](https://doi.org/10.5772/intechopen.112533).
- **Afessa, M.M., Debiagi, P., Ferreiro, A.I., Mendes, M.A.A., Faravelli, T., Ramayya, A.V., 2022.** Experimental and Modeling investigation on pyrolysis of agricultural biomass residues: Khat stem and coffee husk for bio-oil application. J. Anal. Appl. Pyrolysis 162, 105435. <https://doi.org/10.1016/j.jaap.2022.105435>.
- **Afessa, M.M., Ramayya, A.V., Nega, D.T., Mossisa, H.M., Locaspi, A., Faravelli, T.F., 2023.** Investigation of Kinetic, Thermodynamic and Performance Parameters during Devolatilization of Agricultural Biomass Waste via TGA. Ethiop. J. Appl. Sci. Technol. 16–24.
- Caraccio R, Signori L, **Afessa MM**, Caregnato D, Scialabba A, Della TOFFOLA D, et al. CHAR:ME:biochar and biomass-derived waste products as sustainable and safe domestic fuel. In 2024 [cited 2025 Mar 23]. Available from: <https://re.public.polimi.it/handle/11311/1286233>.
- Million M. Afessa, A. Venkata Ramayya, I. Andrea Locaspi, Paulo Debiagi, Alessio Frassoldati, Tiziano Faravelli, AnnaSzepannek, Angela Hofmann, Christoph Pfeifer, Pyrolysis of Large Wood Particles: Kinetic Modeling and Experimental Validation. Joint Meeting the Belgian and Italian Sections of the Combustion Institute, Florence, Italy – 2023.

CERTIFICATIONS & TRAINING

- Key Essentials: The Sustainable Development Goals and the Law – Lucy Cavendish College, University of Cambridge (Sep 2024)
- Organizational Development for Volunteer-Based Initiatives – YALI Edo State (Jul 2024)
- Understanding Renewable Energy – YALI Edo State (Jul 2024)
- Understanding Climate Change – YALI Edo State (Jul 2024)
- Strategies for Personal Growth and Development – YALI Edo State (Jul 2024)
- Teaching Career Readiness – Integrated Youth Activity – Kefeta (Feb 2024)
- Master Class for Teaching Online (Open edX) – EdPlus, Arizona State University (Jun 2024)
- Energy Research Accelerator's Policy Masterclass in Energy Storage – ERA (Mar 2023)
- Global Energy Transition Masterclass 2022 – Endress+Hauser Group (Mar 2022)
- Advanced Analytical Instrumentation Techniques Training (XRD, FTIR, SEM, TEM, VSM, EDX, UV-Vis, DSC, TGA, GC-MS, HPLC, NMR, BET) (Dec 2021)
- Research Writing and Publication Workshop – Aristocrat Research (Jan 2022)
- Institutional IP Policy & TISC Training – Jimma University & Ethiopian Intellectual Property Office (Mar 2021)
- Small Hydro Operation & Maintenance Training – GE Hydro Technical Training (Mar 2018)
- Best Project Award – Energise-Edulink Training – Renewable Energy Electrification, Nairobi (Jul 2016)
- Training on Teaching Methodologies, Research & Industry Partnership – Nairobi (Jul 2016)
- Community Service Recognition – Jimma Institute of Technology (Oct 2016)
- Technical Training (OpenFOAM Software & Soft Skills) – Jimma Institute of Technology & Politecnico di Milano (Mar 2015).

COURSES TAUGHT

- Engineering Thermodynamics I and II, Fluid Mechanics, and Power plant and the Environment
- Advanced Solar Thermal Systems Engineering
- Distributed Energy Systems and Storage
- Advanced Thermodynamics and Energy conversion
- Advanced Thermochemical conversion Technologies (Pyrolysis, Gasification, combustion and etc.)
- Biochemical conversion: anaerobic digestion, anaerobic fermentation, digester design - Biogas, stoves, etc.
- Energy conversion and the Enviroment including Energy audit
- Renewable Energy Technology System Design
- Energy project management and planning

SKILLS (TECHNICAL)

- Curriculum Development & Academic Instruction
- Research Design & Scientific Publishing
- Proposal Writing & Grant Acquisition
- Technical Report Writing
- Project Management in Research
- Kinetic Modeling & Simulation
- Biomass Conversion Technologies

LANGUAGE

English (C1/2 Proficient User)

Listening

Reading

Speaking

Writing

REFERENCES

Tiziano Faravelli (Full Professor),
Department of Chemistry, Materials, and
Chemical Engineering, Politecnico di Milano,
CRECK Modeling Research Group.
Email: tiziano.faravelli@polimi.it
Phone: +390223993282

A.Venkata Ramayya (Full Professor),
Faculty of Mechanical Engineering,
Sustainable Energy Engineering
Jimma Institute of Technology (JiT)
Jimma University, Jimma, Ethiopia
Email: dra.venkata@ju.edu.et
Phone: +251919142891
P.O.Box: 378, Jimma, Ethiopia

SKILLS (DATAANALYSIS)

- Programming & Simulation: MATLAB C+
+, Python, OpenSMOKE++, ANSYS.
- Design & Engineering Software: AutoCAD
SolidWorks, CATIA V5.
- Data Analysis: OriginPro SPSS, Microsoft
Excel.
- Operating Systems: Windows, Ubuntu.

SOFT SKILLS

- Strong communication and presentation
- Analytical and critical thinking
- Effective teamwork and collaboration
- Time management and organizational skills
- Report writing and documentation
- Adaptability and problem-solving

Balewgize A. ZERU (Asst. Prof.)
Dean, Faculty of Mechanical Engineering
Jimma Institute of Technology Jimma University
Email: balewgize.zeru@ju.edu.et
Mobile: +251 931 985926
P.O.Box: 378, Jimma, Ethiopia